When Prediction Transfer Does Not Imply Ranking Transfer: Explicit-Label-Free Learning Across Equity Identities

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Abstract

A return forecast may have low prediction error while providing little guidance about which equity to select from a contemporaneous candidate set. We examine this discrepancy under transfer to equities withheld from model fitting. The evaluation combines training-only calibration of a historical-return baseline, a between-date/within-date squared-error decomposition, same-date ranking measures, and calendar-block inference that respects repeated observations of market dates. An earlier 20-partition analysis suggested a small gain from fitting ridge regression to date-centered outcomes; its prediction-error comparison used an uncalibrated historical-return score, and its partitions shared market history. We revisit those limitations in a separate, post-hoc 2021–2022 evaluation. At the 21-session horizon, calibrated Ret63 has mean absolute error 0.07441, compared with 0.07481 for state ridge and 0.07428 for residual ridge. The large apparent advantage over raw Ret63 therefore disappears after calibration. For seven-, 21-, and 63-session horizons, residual-minus-state mean Spearman differences are -0.0109, 0.0400, and -0.0028; *none of the three horizons shows a statistically supported ranking advantage under the reported dependence-aware sensitivity analysis.* All three-month circular-block percentile intervals contain zero, with only 24 unique evaluation months. A positive seven-session time–identity interaction is exploratory and does not withstand a minimal correction across three horizons. The results demonstrate how calibration, decision-aligned scoring, and the choice of inferential unit change the interpretation of apparent forecast transfer. The fixed historical cohort and short evaluation period limit generalization.

Keywords: cross-sectional prediction; ranking transfer; unseen equities; forecast calibration; block bootstrap; evaluation design.

1 Introduction

When an investor selects an equity from a set of eligible candidates, the immediate decision is comparative: which security should receive the next allocation? Regression models are commonly trained on pooled asset–time observations using squared or absolute prediction error. Those losses assess the accuracy of return estimates, while the selection depends on how the estimates order securities on the same date. Good pooled forecasts may therefore provide weak guidance for a cross-sectional decision.

Transfer to new securities raises an additional issue. A model trained without ticker or company-name fields can still exploit persistent characteristics that distinguish assets. Here the six numerical predictors include historical returns, volatility, drawdown and a market return; they may collectively encode features associated with particular firms. We use *explicit-label-free* to mean that no categorical asset label enters the predictor. It does not imply that the numerical states are independent of identity or that a rule is unaffected by the composition of the investable universe.

An earlier group-holdout analysis found low return-prediction errors for state-based ridge models but weak same-date ranking. Its 20 fixed identity partitions yielded a residual-target versus ordinary ridge mean Spearman difference of 0.003656, with a reported partition-based 95% interval $[0.001359, 0.005952]$. Strict evaluation on later dates and unseen securities remained unfavorable. These findings exposed two evaluation concerns. First, the original level-error comparison treated raw historical 63-session return, a directional ranking heuristic, as though it were already calibrated to predict a clipped forward benchmark-relative return. Second, the fixed identity partitions reused the same calendar dates and many of the same stocks. Their apparent number of replications overstated the independent time information available for inference.

We examine these concerns with a common evaluation procedure. A training-only affine calibration puts Ret63 on the forecast target's scale, while the raw score remains a separate momentum ranking comparator. For any fixed cross-section, squared forecast error is decomposed exactly into errors in the date mean and in the within-date residuals. We compute same-date Spearman correlation and top-decile outcomes for each model and construct paired monthly contrasts before applying circular calendar-block resampling. A separate fixed-roster design compares model differences over near and future dates and over identities present or absent from fitting.

The components serve distinct purposes. Calibration isolates the effect of forecast scale. Date-centered metrics measure the ordering relevant to the allocation choice. Monthly aggregation prevents repeated identity partitions from being treated as independent market realizations. The time–identity comparison describes transfer variation without assigning a causal interpretation to market-period or stock-group membership. Together they form a reproducible evaluation procedure for checking whether gains in numerical forecasting have decision-relevant support.

The results alter the original comparison. Calibrated Ret63 is close to the two ridge predictors on mean absolute error, and the residual-target model's ranking difference changes sign across seven, 21 and 63 trading sessions. None of the three horizons has a three-month calendar-block interval that excludes zero. The tests use only 24 distinct future evaluation months. Because the evaluation design followed examination of the earlier findings, the new tests are post-hoc sensitivity analyses; they are not independent, preregistered confirmation. The contribution is an empirical audit of forecasting evaluation under identity transfer, together with explicit definitions of the statistical objects on which that audit rests.

2 Related Work

Forecasting returns across equities. Machine-learning asset-pricing studies estimate risk premia from firm characteristics and document the role of nonlinear relations in return forecasting (Gu et al., 2020). Here a small, fixed ridge family makes the evaluation question more transparent. The issue is whether forecasts trained on one group of securities retain accuracy and cross-sectional usefulness when applied to withheld identities. A group holdout evaluates a different form of generalization from a chronological split that reuses familiar securities.

Forecasts and selection decisions. Cross-sectional equity selection uses relative scores among assets available at the same decision date. A level forecast can move every predicted return in the same direction without altering which equity ranks first. Learning-to-rank research therefore treats the ordering itself as a target (Poh et al., 2021); decision-focused learning distinguishes prediction loss from the quality of decisions made with a forecast (Elmachtoub and Grigas, 2022). This distinction is particularly consequential in equity allocation, where capital goes to a subset of the predicted winners even if the predicted common market level is accurate. Our residual-target fit holds the features and model family fixed, allowing the effect of one alternative regression target to be inspected alongside numerical and ranking outcomes. It does not solve the portfolio-allocation optimization problem.

Baseline calibration and model comparison. Simple allocation rules can remain competitive when estimated parameters are noisy (DeMiguel et al., 2009). The raw Ret63 score in this study is a positive-momentum ranking rule. For MAE and MSE comparisons we also fit a training-only map from that score to the forward return target. Reporting both versions keeps the original directional rule distinct from the regression-based forecast, whose fitted slope may reverse the ranking. Specification search and data snooping can distort financial backtests (White, 2000; Arnott et al., 2019; Bailey et al., 2014; Harvey et al., 2016); inspecting an earlier result before designing new checks further limits the claims supported by the new evidence.

Inference with a shared market calendar. Models evaluated on different stocks but the same dates experience common market conditions. Resampling individual observations, or treating repeated identity partitions as independent replications, overlooks that dependence. Block resampling addresses part of the problem under assumptions about the underlying temporal process (Künsch, 1989; Politis and Romano, 1994). We use fixed-length circular blocks of complete calendar months, rather than the stationary bootstrap in Politis and Romano (1994). With 24 future months, the intervals are reported as sensitivity analyses rather than as guaranteed small-sample coverage.

3 Problem Setting and Information Structure

3.1 Asset–time observations and the outcome clock

At monthly decision date t, let $\mathcal{A}_t$ denote the equities retained by the supplied panel's eligibility and availability rules. An observation is $(i, t, X_{i,t}, Y_{i,t}^{(H)})$, where i indexes the execution and data-joining identity, $X_{i,t} \in \mathbb{R}^6$ is the observed numerical state, and $H \in \{7, 21, 63\}$ counts future trading sessions. The six predictors are one-, 21- and 63-session historical returns, 63-session historical volatility, distance from a 252-session drawdown reference, and the 63-session SPY

return. No categorical ticker or company label enters the main ridge predictor. The benchmark for the forward target is QQQ, not SPY.

For the supplied price construction, historical signals are calculated through the previous trading session's close, followed by an entry at decision-date adjusted open. The benchmark-relative target is the clipped forward log-return difference,

$$Y_{i,t}^{(H)} = \mathrm{clip}_{[-1,1]}\left[\log\left(\frac{P_{i,t+H}^{\mathrm{adj,cl}}}{P_{i,t}^{\mathrm{adj,op}}}\right) - \log\left(\frac{P_{Q,t+H}^{\mathrm{adj,cl}}}{P_{Q,t}^{\mathrm{adj,op}}}\right)\right], \tag{1}$$

where Q denotes QQQ and $t+H$ means the Hth future session in the supplied trading calendar. This is a research reward proxy. It is neither a transaction-cost-adjusted portfolio return nor an estimate of an investable strategy's utility.

Let $\mathcal{F}_{t^-}$ denote information available before the decision-date open. Our chronological requirement is that $X_{i,t}$ and the eligibility decision be measurable with respect to $\mathcal{F}_{t^-}$. The future label only enters fitting after its maturity date $m_H(t) \le T_0$, where T_0 is the training information cutoff. In the new code, the fitting sample for any identity partition is

$$\mathcal{D}_{\mathrm{tr}}(T_0, I_{\mathrm{tr}}) = \{(i,t) : i \in I_{\mathrm{tr}},\ t \le T_0,\ m_H(t) \le T_0\}. \tag{2}$$

Standardization, ridge coefficients and training-date target centering are fitted on this set alone.

3.2 Prediction and ordering are different evaluation objects

For numeric predictions $s_{i,t}$, pooled squared error is $N^{-1}\sum_{t,i}(Y_{i,t} - s_{i,t})^2$, with $N = \sum_t |\mathcal{A}_t|$. Its dependence on scale makes it appropriate for forecasts whose units match those of Y. To assess relative ordering we first calculate $\rho_{\mathrm{S}}(\{s_{i,t}\}_{i\in\mathcal{A}_t}, \{Y_{i,t}\}_{i\in\mathcal{A}_t})$ within each eligible cross-section and average these date-level values. We also report top-decile lift, the realized outcome mean among the highest-scored decile minus the same-date eligible-asset mean, and top-one regret, the gap between the highest realized outcome and the outcome of the highest-scored asset. A top-decile lift need not share the sign of mean Spearman because the former tests one selected tail, whereas the latter summarizes the full cross-sectional ordering.

The set $\mathcal{A}_t$ remains part of the estimand. Relabeling a fixed set cannot change an explicitly label-free scoring rule if all tie resolution is label-neutral, but adding or deleting candidates can change an arg max choice. Hence neither explicit-label omission nor good performance on one fixed universe proves invariance to universe composition.

4 Models and Controlled Comparators

4.1 Pooled state ridge

For each training set in (2), a StandardScaler is fitted using only the training rows. Let $\widetilde{X}_{i,t}$ include a fitted intercept and the standardized numerical state. State ridge solves

$$\widehat{\theta}^{\mathrm{state}} = \arg\min_{\theta}\left\{\sum_{(i,t)\in\mathcal{D}_{\mathrm{tr}}}\left(Y_{i,t}^{(H)} - \theta^\top \widetilde{X}_{i,t}\right)^2 + \lambda\|\theta_{\mathrm{slope}}\|_2^2\right\}, \tag{3}$$

with the intercept unpenalized by the software implementation. The software configuration fixes $\lambda = 5$ in the V2 audit. The same fitted scaler and coefficients are used for all held-out outcomes;

no information from the evaluation dates is used to refit them.

4.2 Ridge on cross-sectional residual outcomes

For a matured *training* date, write

$$\overline{Y}_t^{\mathrm{tr}} = |I_t^{\mathrm{tr}}|^{-1} \sum_{i \in I_t^{\mathrm{tr}}} Y_{i,t}^{(H)}, \qquad \widetilde{Y}_{i,t}^{\mathrm{tr}} = Y_{i,t}^{(H)} - \overline{Y}_t^{\mathrm{tr}}, \tag{4}$$

where I_t^{tr} includes only eligible training identities with fully matured labels. CS-residual ridge uses the same learner family, numerical states, and regularization penalty as (3) but replaces Y with $\widetilde{Y}^{\mathrm{tr}}$ in the loss. At deployment there is no test-date outcome centering, since the test-date outcomes do not exist when scores are formed. The score is the model's predicted training-residual target. Residualization changes a supervised training objective and does not produce an empirical guarantee of rank improvement.

4.3 Raw and calibrated 63-session return

A 63-session historical return is a simple momentum-style ranking score. Calling that raw score a point prediction of future QQQ-relative log return exposes it to scale and intercept mismatch: the forward target in (1) is bounded and benchmark-relative, while the raw return has neither adjustment. We retain two distinct comparisons.

Raw Ret63 assigns scores $s_{i,t}^{\mathrm{raw}} = r_{i,t}^{(63)}$ in the original positive-momentum direction. It is principally a ranking baseline. *Calibrated Ret63* predicts the forward target using the fitted affine map

$$\widehat{Y}_{i,t}^{\mathrm{cal}} = a_k + b_k r_{i,t}^{(63)}, \tag{5}$$

where k is the identity partition and (a_k, b_k) are the intercept and raw-unit slope implied by a training-only StandardScaler plus ridge regression on the same matured forward label as the state model. The $\lambda = 5$ penalty is fixed rather than chosen on test performance. This is a fairer predictor-scale comparison, although calibration does not equalize all modeling choices or training flexibility.

A negative b_k reverses the ranking of $r^{(63)}$ whenever observations are non-tied. In that case the calibrated baseline is a reversal signal, not the original positive-momentum heuristic. We report both, rather than choosing the more favorable direction after observing evaluation labels. Appendix A.5 records the raw-unit conversion and the full ranges of fitted coefficients. The distinction is essential: improved MAE from an affine rescaling is not evidence that the original momentum trading rule has improved.

4.4 Identity diagnostics

The archived study also included an identity reconstruction task and a limited regularized identity-aware regression. Their role is diagnostic. A classifier's ability to infer a ticker from numerical states implies that labels are not statistically erased; conversely, failure of a specific dummy-augmented model to improve a reported error does not establish $\mathbb{E}[Y \mid X, I] = \mathbb{E}[Y \mid X]$. The two prediction-loss comparisons in the archived ablation have different recorded sample sizes ($n = 338$ and $n = 334$), limiting their interpretation as an exact paired controlled experiment. The V2 audit does not refit this ablation, and we do not promote it to a new validated result.

5 Theoretical Separation of the Evaluation Targets

The statements here identify consequences of the statistical objectives. They are not claimed as newly discovered probabilistic principles. Proofs and qualification of the stochastic example appear in Appendix A.

Assumption 1 (Conditional-mean sufficiency for a scalar reward)**.** *For integrable Y, observed numerical state X, and explicit identity I, suppose $\mathbb{E}[Y \mid X, I] = \mathbb{E}[Y \mid X]$ almost surely under the target law.*

Proposition 1 (Identity redundancy for candidatewise conditional-mean scoring)**.** *Under Assumption 1, consider the restricted decision class that assigns to each candidate i a score based on its* own *information (X_i, I_i), with the common target-law conditional mean as the score and an identity-independent tie rule. Within this candidatewise class, the conditional-mean ranking can be implemented using X_i alone, up to ties.*

This statement is deliberately restricted. It does *not* assert that the same assumption makes asset labels redundant for a Bayes decision conditioning jointly on the entire candidate set $(X_1, I_1, \ldots, X_n, I_n)$; the information of other candidates can change the conditional reward of candidate i. Sufficient joint-information conditions would have to be stated separately. Assumption 1 is weaker than conditional independence of the entire outcome distribution. Neither label permutation nor identity-reconstruction accuracy provides an empirical test sufficient to verify it, and transfer through calendar distribution shift requires additional assumptions.

Proposition 2 (Exact between-date/within-date error decomposition)**.** *For any date with n_t eligible assets, let $\bar{Y}_t$ and $\bar{s}_t$ be the outcome and prediction means, and define $R_{i,t} = Y_{i,t} - \bar{Y}_t$ and $Q_{i,t} = s_{i,t} - \bar{s}_t$. Then*

$$\frac{1}{n_t} \sum_{i \in \mathcal{A}_t} (Y_{i,t} - s_{i,t})^2 = (\bar{Y}_t - \bar{s}_t)^2 + \frac{1}{n_t} \sum_{i \in \mathcal{A}_t} (R_{i,t} - Q_{i,t})^2. \tag{6}$$

Consequently pooled MSE is the n_t/N-weighted average of the two terms.

The first term can be reduced by capturing a date-common level even if the second term is nearly untouched. For Spearman, the common level cancels from the within-date ordering. The decomposition is a squared-error identity; there is no corresponding additive decomposition of MAE.

Theorem 1 (Low prediction error need not provide ranking information)**.** *For a cross-section containing $n \geq 3$ assets, there are nonconstant continuous-valued predictions and outcomes such that their expected same-date Spearman correlation is zero, while their expected squared prediction error can be made arbitrarily small.*

The construction uses mutually independent continuous relative components of vanishing scale, together with an exactly available date-common component. It demonstrates logical non-implication. In particular it avoids assigning numerical Spearman zero to a cross-section with constant predictions, for which Spearman is undefined. A small MSE achieved by predicting an almost constant target should not, without additional rank assumptions, be interpreted as successful asset ordering.

Proposition 3 (Date centering preserves the observed ordering)**.** *For every date and two candidates i, j, subtracting the same realized cross-sectional mean preserves their pairwise ordering:*

$$Y_{i,t} > Y_{j,t} \quad \Longleftrightarrow \quad Y_{i,t} - \bar{Y}_t > Y_{j,t} - \bar{Y}_t. \tag{7}$$

Table 1: Evidence protocols. Rows describe different estimands and should not be pooled into a single confirmation claim.

	Historical experiment	V2 post-hoc audit
Emphasis	Identity holdout and strict double-OOS boundary	Calibration, loss decomposition, date dependence, fixed-roster interaction
Repeated groups	Five development; 20 later fixed partitions	20 overlapping partitions in held-out analysis
Calendar protocol	Full-history identity holdout; separate 2021/22/23 cutoffs	Pre-2021 fit; 2021–22 test (24 months)
Predictor comparison	State ridge, residual ridge, raw Ret63 heuristic	Same ridge family plus calibrated Ret63 and raw Ret63
Inference status	Reported split-level paired intervals	Calendar-block nominal sensitivity; post-hoc

This proposition concerns the target ordering, not the trained ridge coefficients. Training with centered labels can change coefficient estimates and their out-of-sample ranks in either direction, as the empirical results below illustrate.

Lemma 1 (Candidate-set dependence)**.** *For any score function $s(X)$, adding a candidate whose score exceeds the current maximum changes a highest-score selection rule, even when all retained scores are unchanged.*

Thus explicitly label-free scoring is not universe-independent. Missing delisted or historically unavailable firms can influence both which candidates enter a dated cross-section and which asset is selected.

6 Empirical Protocol and Evidence Hierarchy

6.1 Archival identity experiment and V2 evaluation audit

The initial research had five development identity holdouts, subsequently followed by 20 fixed identity partitions, horizon checks at $H = 7, 21, 63$, four frozen-universe perturbations and three strict future-time $\times$ unseen-identity cutoffs (2021-01-04, 2022-01-03 and 2023-01-03). The later 20 partitions were frozen for that historical analysis after the residual-learning hypothesis emerged in development. Their overlaps in asset records and market dates prevent treating 20 partition-level statistics as 20 independent replications of calendar experience.

V2 is a *different* post-hoc experiment, not a rerun of the same archival inference with only a different standard error. The core evaluation freezes a 2020-12-31 training-information cutoff and holds out 2021-01-04 through 2022-12-30, with 20 overlapping but deterministically reproducible identity partitions and approximately 40 held-out identities per partition. For $H = 21$, eligible training signals end on 2020-12-01 while their final outcomes mature by 2020-12-31. The main test has 24 distinct calendar months, each scored repeatedly under the partition protocol. All models, baselines and normalization parameters are fit before the test period.

6.2 Data, information audit and known external validity limits

The supplied V2 panels contain 22,063 records (245 identities, 105 observation dates) at $H = 7$, 21,818 (244, 104) at $H = 21$, and 21,331 (241, 102) at $H = 63$, spanning decision dates from 2018 into August 2026. These are panel-wide totals, not the count of independent test months. The accompanying software audit reports agreement between the six features and forward labels

and recalculation from the archived adjusted price files across 65,212 panel rows to floating-point precision. The 16 software tests, maturity filters and identity-disjointness checks passed during the V2 run.

These checks establish consistency with the *supplied adjusted-price archive*. They cannot establish that the historical adjustment factors were available at the recorded decision time, that removed or delisted firms were represented, or that the stock cohort satisfies a full historical point-in-time security-master definition. Our evidence consequently applies to this fixed engineering cohort, not the complete investable US equity population. The clipped relative target also does not reflect spread, commissions, market impact, corporate-accounting frictions or portfolio compounding.

6.3 Metrics and paired construction

The main metrics are MAE, MSE, same-date Spearman, top-decile lift and top-one regret. We also separate date-common squared loss and within-date squared loss using Proposition 2. Comparisons of two learned models use the *same* held-out rows and score dates within each partition. No individual asset–time row is treated as an independent ranking task. Training parameters, calendar boundaries, seeds and cohort membership are fixed in the V2 verification archive. The full model-refitting software and raw market source archive are separate from this manuscript's plotting-and-compilation bundle.

6.4 Use of generative AI in research support

OpenAI ChatGPT (including GPT-6; used during 2026) assisted with the formulation and exposition of evaluation diagnostics, drafting and review of Python analysis and test code, and interpretation checks against saved numerical outputs. The human authors directed the research questions and evaluation choices, ran the experiments and software tests, and assessed the resulting calculations and interpretations. AI-generated text was not treated as market data or as independent empirical evidence. The authors retain responsibility for the accuracy of the methods, software, numerical reporting, and conclusions.

7 Empirical Results: Forecast Scale and Cross-Sectional Ordering

7.1 What the archival identity holdouts showed

The original 20-partition experiment reported mean same-date Spearman correlations of 0.00113 for state ridge, 0.00479 for CS-residual ridge and 0.01384 for the raw Ret63 ranking score. The paired residual-minus-state Spearman difference was 0.003656; the reported partition-based 95% interval was $[0.001359, 0.005952]$. Top-decile lift increased by 0.003319 under the same calculation. The level-prediction MAEs were 0.08336 for either ridge score and 0.20200 for *raw* Ret63. Because that last quantity is a historical ranking score, its MAE does not establish the size of the ridge advantage relative to a calibrated forecast.

The archived strict double-OOS analysis gives a less favorable picture of rank transfer. Averaging the three reported time cutoffs yielded Spearman approximately -0.081 for state ridge, -0.044 for CS-residual ridge and -0.010 for raw Ret63. These are not observations generated by the V2 code. We retain them as a boundary result rather than replacing them with the later protocol's different future-window numbers. The archived identity reconstruction accuracy was

Table 2: V2 held-out prediction errors. Entries are averaged across the fixed, repeated identity protocol on the indicated 2021–2022 evaluation windows. Calibrated Ret63 is on the forward-target scale; raw Ret63 is reported only to expose the scale mismatch. Smaller is better.

Model	$H = 7$ MAE	$H = 21$ MAE	$H = 63$ MAE
State ridge	0.048595	0.074811	0.128523
CS-residual ridge	0.048617	**0.074280**	0.127762
Ret63, trained affine calibration	**0.048438**	0.074408	**0.127271**
Ret63, raw uncalibrated score	0.158117	0.168057	0.202453

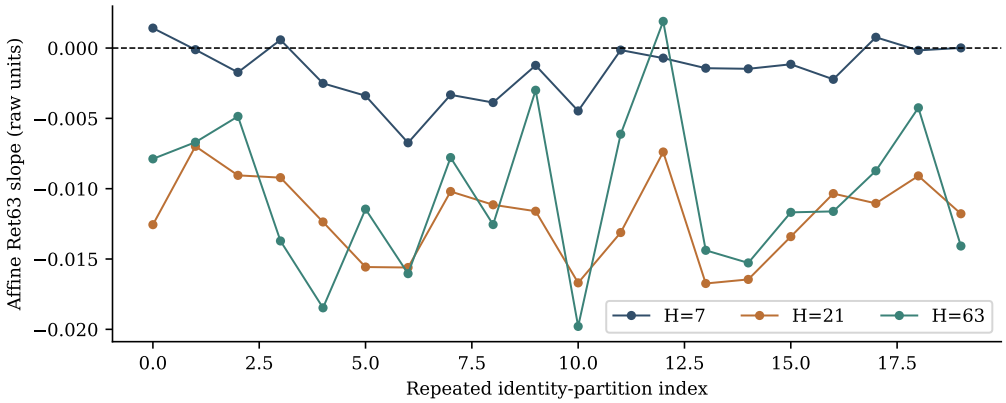


Figure 1: Training-only raw-unit affine Ret63 slopes by identity partition. Every 21-session slope is negative. A slope below zero reverses ranking relative to the uncalibrated momentum rule; the plot does not establish stable future mean reversion.

0.13254 across 40 identities against uniform-chance accuracy 0.025, consistent with numerical features containing an identity fingerprint despite removal of explicit labels.

7.2 Forecast-scale comparison in the V2 protocol

Table 2 changes the interpretation of the prediction comparison. For 21 sessions, calibrated Ret63 has MAE 0.074408, against 0.074811 for state ridge and 0.074280 for residual ridge. The previously striking ridge-versus-Ret63 gap largely disappears when the simple historical return is fitted to the forward outcome's scale. On the same protocol state ridge minus calibrated Ret63 has a mean date-paired MAE difference +0.0004022. In exploratory three- and six-month blocks the associated two-sided centered-bootstrap p values are 0.0400 and 0.0206, respectively; the time sample and specification-search limitations discussed below preclude a broader superiority claim. At seven and 63 sessions calibrated Ret63 has the smallest observed MAE among the three calibrated estimators. This is a conditional comparison among deliberately restricted methods, not evidence that historical returns are the best feasible forecast.

The raw versus calibrated distinction is not cosmetic. In the 21-session held-out experiment all 20 independently fitted affine calibrations have *negative* slopes between −0.016740 and −0.006986, with intercepts between approximately −0.004310 and −0.002115. Their rank order is thus the reverse of raw Ret63. At $H = 7$ and $H = 63$, respectively 16 and 19 of the 20 slopes are negative. A data-driven prediction fit need not implement the raw momentum rule; calling both procedures "Ret63" without distinguishing their orientation could conceal the change in action ranking. Figure 1 shows all slopes, not a selected favorable partition.

Table 3: V2 mean same-date Spearman rank correlations. Positive values indicate positive ordering on average; the values are point estimates, not significance tests.

Model	$H = 7$	$H = 21$	$H = 63$
State ridge	−0.032887	−0.087189	−0.089241
CS-residual ridge	−0.043806	−0.047187	−0.092081
Ret63, calibrated	−0.005768	−0.017511	+0.020583
Ret63, raw positive momentum	+0.023539	+0.017511	−0.020167

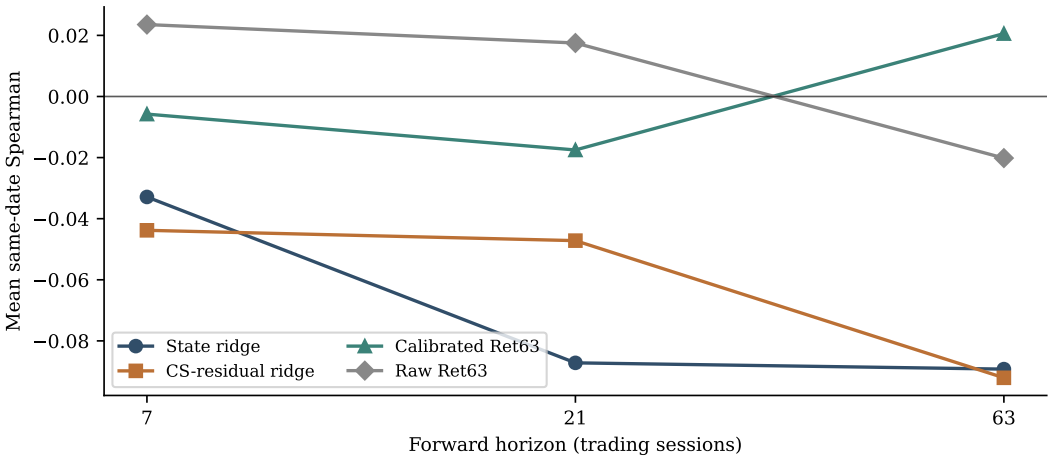


Figure 2: Absolute date-averaged Spearman for the four V2 scores. Calibration can reverse the raw Ret63 ordering; all estimated correlations are small in magnitude. The three horizon panels need not contain identical assets or observation dates.

7.3 Absolute ranking remains weak and horizon-dependent

The residual-minus-state rank differences are −0.010919, +0.040003 and −0.002840 at seven, 21 and 63 sessions. Residual training therefore cannot be summarized as a uniformly rank-improving choice. For 63 sessions the residual learner improves top-decile lift from −0.049458 to −0.036118 even as its mean Spearman becomes slightly more negative. Concentration on the upper tail and full-rank correlation need not agree. Its 63-session MAE is also lower than state ridge's despite the weaker Spearman. These metric disagreements follow the distinction between learning a numerical target and ranking all contemporaneous candidates; they do not imply that a profitable investment action has been identified.

7.4 What the exact squared-error decomposition shows

At 21 sessions, state-ridge pooled MSE is 0.010738, comprising within-date MSE 0.009947 and between-date MSE 0.000791. Corresponding values are $0.010644 = 0.009901 + 0.000743$ for CS-residual ridge and $0.010628 = 0.009824 + 0.000804$ for calibrated Ret63. The decomposition in Proposition 2 is exact for each matched group of scored assets and remains exact under pooled averaging with the appropriate weights. In this evaluated window most measured squared error lies in the within-date component. Thus the generic mathematical argument that one *can* achieve low MSE by predicting a common component does not establish that common-component accuracy is the dominant mechanism in this particular V2 sample. The evidence supports an objective distinction; it does not identify a unique causal mechanism for the model rankings.

Table 4: V2 CS-residual minus state ridge, with the sign interpreted separately for each metric. These are within-protocol point estimates.

Metric	$H = 7$	$H = 21$	$H = 63$
Mean Spearman	−0.010919	+0.040003	−0.002840
Top-decile lift	−0.004306	+0.004351	+0.013340
MAE (negative favors residual ridge)	+0.000021	−0.000530	−0.000761

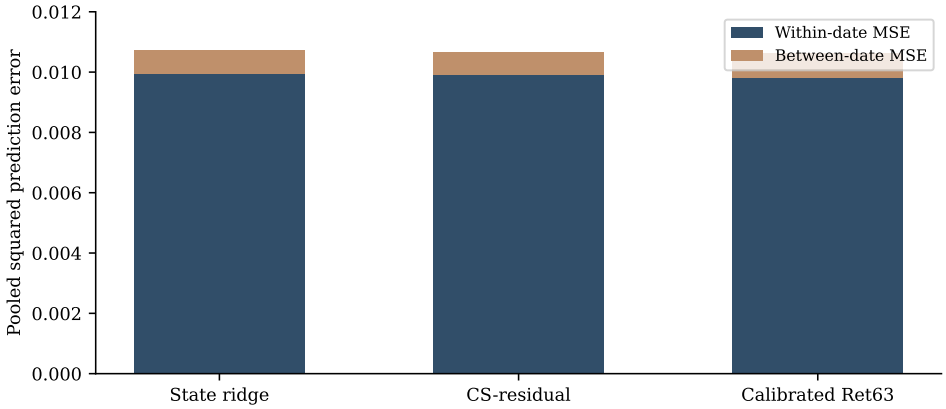


Figure 3: V2 21-session pooled MSE decomposed into within-date and between-date contributions for the three calibrated predictors. The identity is mechanical; component magnitudes are empirical.

8 Temporal Dependence and the Evidence for Rank Improvement

8.1 The resampling object is a calendar month

Let $m_{k,t}^{(A)}$ and $m_{k,t}^{(B)}$ be a metric evaluated for two models on identical held-out candidates in partition k and date t. The paired contrast $d_{k,t} = m_{k,t}^{(A)} - m_{k,t}^{(B)}$ is averaged across the $K = 20$ repeated partitions at each test month, $\bar{d}_t = K^{-1} \sum_k d_{k,t}$. Inference then concerns the average of the 24 monthly $\bar{d}_t$ values. The asset-time rows and the 20 partitions are not 480 independent market experiments. This construction retains comparisons on matched dates and uses the calendar month as the resampled time index. It does not, by itself, eliminate all dependence from repeated asset appearances or imperfect temporal stationarity.

The V2 code draws $B = 5{,}000$ fixed-length circular blocks of three, six or 12 months with replacement and forms nominal two-sided 95% percentile intervals from the empirical distribution of resampled mean contrasts. To produce an exploratory two-sided p value, it centers $\bar{d}_t$ under the mean-zero null, repeats the block bootstrap and counts how often the absolute centered resampled mean is at least the observed absolute mean, with +1 correction in numerator and denominator. Such p values are neither exact finite-sample tests nor guaranteed to be calibrated in the present financial time series. A count of zero exceedances produces a lower reporting bound $1/(5001)$, not $p = 0$.

8.2 Results and block-length sensitivity

The three-month exploratory centered-bootstrap p values are 0.4109, 0.1458 and 0.9314 for $H = 7, 21, 63$. All nine intervals across the three block lengths contain zero. The 21-session point estimate is positive, but these sensitivity analyses do not establish a stable nonzero ranking effect. Neither does a confidence interval containing zero prove that the two models

Table 5: V2 CS-residual minus state-ridge same-date Spearman, paired date-block sensitivity. All intervals are nominal 95% percentile intervals. Three-month p values are exploratory; 12-month intervals are descriptive only.

H	Mean $\Delta\rho$	3-month interval	6-month interval	12-month interval
7	−0.010919	[−0.039495, 0.014636]	[−0.034061, 0.008927]	[−0.028400, 0.006562]
21	+0.040003	[−0.012674, 0.094728]	[−0.014641, 0.100090]	[−0.006492, 0.084879]
63	−0.002840	[−0.059347, 0.057459]	[−0.043324, 0.035333]	[−0.037793, 0.032605]

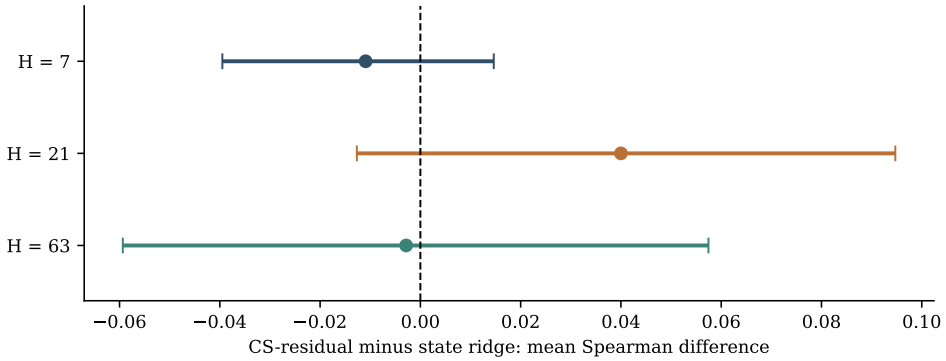


Figure 4: Mean CS-residual minus state-ridge Spearman and nominal 95% circular three-month calendar-block intervals, using 5,000 resamples. All intervals cover zero.

are equivalent. With only 24 months, the nominal counts of nonoverlapping blocks are eight, four and two for the three lengths, respectively. In the 12-month case there is insufficient temporal replication for meaningful inferential interpretation. The V2 output labels that case `descriptive_only_too_few_blocks` and suppresses an inferential p value.

A nuance concerns other metrics. At 21 sessions, the three-month nominal percentile interval for CS-residual minus state-ridge MAE is $[-0.001097, -0.000037]$, yet the separately centered two-sided bootstrap returns $p = 0.0518$. These calculations are not inversions of one common pivotal test and need not agree exactly at a decision threshold. We do not select the favorable interpretation. We prioritize the effect size, estimation target and dependence qualification, and regard the inferential discrepancy as a reason not to make an unconditional significance claim for this secondary outcome.

8.3 Comparison with the historical 20-partition interval

The historical 20-partition Spearman improvement of +0.003656 and its narrow interval pertain to a different evaluation protocol from V2's +0.040003 21-session estimate. The V2 interval cannot be understood as a dependence correction applied to the same point estimate. Both observations are reported with their source and design. Jointly they show that a small apparent improvement is sensitive to how identity groups and calendar periods are sampled; they do not isolate which one of these differences explains the change in magnitude. A date-block sensitivity result on only 24 months does not supersede all historical evidence, but it materially narrows the claim that the residual objective improves ranking reliably.

Table 6: V2 21-session two-by-two diagnostic. Same-date Spearman is evaluated within each group's own contemporaneous cross-section. The row contrast is CS-residual minus state ridge.

Time × roster	State Spearman	CS Spearman	CS − State
Near × seen	+0.068410	+0.046824	−0.021585
Near × unseen	+0.088964	+0.043939	−0.045025
Future × seen	−0.004363	+0.033186	+0.037549
Future × unseen	−0.035671	+0.020478	+0.056148

9 Time–Identity Evaluation: An Associational Interaction

9.1 Definition and interpretation

The separate V2 two-by-two diagnostic trains the predictors before 2020 and evaluates one fixed identity split in a *near* regime (12 months during 2020) and a *future* regime (24 months during 2021–2022). Identity members observed in the training roster are called seen; withheld identities are called unseen. For metric M, define on the same calendar date t the model contrast

$$\delta_{t,g} = M_{t,g}(\text{CS residual}) - M_{t,g}(\text{state}), \qquad g \in \{\text{seen}, \text{unseen}\}. \tag{8}$$

Pair the identity-group differences within each date as $G_t = \delta_{t,\text{unseen}} - \delta_{t,\text{seen}}$. For horizon H the reported interaction is

$$\widehat{\Gamma}_H = \frac{1}{|\mathcal{T}_{\text{future}}|} \sum_{t \in \mathcal{T}_{\text{future}}} G_t - \frac{1}{|\mathcal{T}_{\text{near}}|} \sum_{t \in \mathcal{T}_{\text{near}}} G_t. \tag{9}$$

The metric here is same-date Spearman. This is a comparison of model-by-roster differences across two historical periods. It is *not* a causal difference-in-differences estimand: neither identity group nor period is assigned randomly, the two rosters have different stocks and cross-sectional sizes, and the relevant parallel-trends interpretation is not assumed. Nor is one group of formerly withheld identities equivalent to all future IPOs or delistings.

9.2 Observed cells and interaction results

The 21-session near-seen, near-unseen, future-seen and future-unseen cells contain, respectively, 1,788, 444, 3,576 and 888 asset–month rows. The near period has 12 months and the future period 24; the seen and unseen groups have 149 and 37 stocks. The future-unseen residual model's Spearman estimate is +0.02048 in this specific diagnostic, whereas it was negative in the historical strict time-cutoff analysis. The two experiments evaluate different rosters and chronological protocols. A positive subgroup value does not repair the negative historical result or establish portability to a broad target population.

The interaction point estimates are +0.015197, +0.042039 and −0.026504 for $H = 7, 21, 63$. V2 resamples full calendar months in fixed-length circular blocks *separately within the near and future regimes*, retaining same-date seen/unseen pairing. This procedure assumes that dependence across the boundary between regimes is negligible enough for the stated sensitivity analysis; the assumption is not empirically verified. The three-month result for $H = 7$ has nominal interval $[0.002814, 0.028045]$ and exploratory $p = 0.0192$. For $H = 21$ and $H = 63$, intervals $[-0.029457, 0.108915]$ and $[-0.072547, 0.018334]$ include zero, with exploratory $p = 0.2366$ and 0.2743, respectively.

Table 7: Fixed-roster time–identity interaction under three-month circular-block resampling ($B = 5{,}000$). The intervals and p values are exploratory.

Horizon	$\widehat{\Gamma}_H$	Nominal 95% interval	Exploratory p
$H = 7$	$+0.015197$	$[+0.002814, +0.028045]$	0.0192
$H = 21$	$+0.042039$	$[-0.029457, +0.108915]$	0.2366
$H = 63$	-0.026504	$[-0.072547, +0.018334]$	0.2743

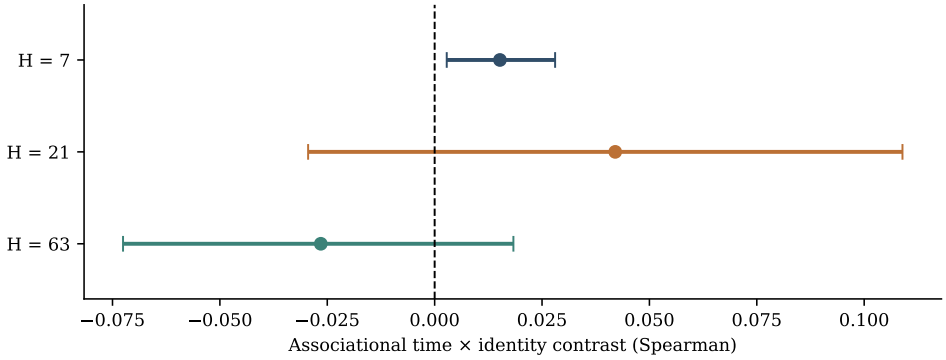


Figure 5: Fixed-roster associational interaction estimates and three-month block percentile intervals. One seven-session interval excludes zero, but there is no consistent sign or replicated evidence across horizons.

9.3 Multiplicity and the limits of the short near window

The near regime supplies only 12 distinct months, or four nominal three-month blocks. Six-month and 12-month near-regime block counts are two and one, respectively; the corresponding longer-block intervals are recorded for descriptive sensitivity and not interpreted as significance evidence. Considering the three horizon comparisons alone, a Bonferroni adjustment maps the smallest exploratory p value to $\min(1, 3 \times 0.019196) \simeq 0.0576$. The actual inquiry spans additional metrics and block sizes, so treating the smallest unadjusted p value as a confirmatory discovery would be unwarranted. Sign changes across horizons are also not themselves a formal test of a horizon-by-interaction difference; a paired cross-horizon inference would require matching their evaluation populations and dependence structures.

10 Discussion, Limitations, and Implications

Calibration changes the forecast-error comparison substantially. Raw historical Ret63 is on a different scale from a clipped QQQ-relative forward log return. After a training-only affine fit, its MAE is comparable to that of the state models; in the 21-session test it is lower than state ridge's and slightly higher than residual ridge's. The fitted slope is negative for all 20 partitions at that horizon. This turns the calibrated forecast into a reversal ranking, while the raw score retains its positive-momentum interpretation. Neither comparison should be substituted for the other.

The residual-target experiment illustrates another difference between forecasting and selection. Centering each training date's outcomes preserves the realized order of those outcomes. Re-estimating ridge coefficients on the centered target may nevertheless change the predicted order in either direction. The model reduces 63-session MAE and improves top-decile lift relative to state ridge, although its full-cross-section Spearman is slightly lower. The within-date component also accounts for most observed squared error in the 21-session window. Thus the data do not

identify improved estimation of a date-common level as the dominant mechanism behind the observed prediction error. The squared-error decomposition establishes what can be separated, while the data determine the size of each component.

The historical 20-partition interval and the new month-block intervals address different evaluation samples. The former was calculated across overlapping identity partitions; the latter first aggregates repeated comparisons by date within a frozen 2021–2022 window. All three horizons have nominal block intervals containing zero for the residual-minus-state ranking difference. These results constrain the strength of the original improvement claim, although they cannot identify whether calendar window, population composition or uncertainty calculation accounts for the different point estimates. The new design was chosen after inspection of the historical evidence and is properly treated as post-hoc.

Statistical limitations. Only 24 future months are available for the principal comparisons and 12 near months for the fixed-roster interaction. The 63-session outcomes overlap in calendar time, and variation in market conditions may persist beyond a selected bootstrap block. Twelve-month blocks leave only two nominal units in the main test and one in the near period. Neither 5,000 resamples nor an interval that happens to exclude zero under a longer block supplies additional historical replication. The seven-session interaction's exploratory p value loses its nominal five-percent status under a minimal correction for the three horizon comparisons. These restrictions preclude a reliable positive interaction claim.

Data and target limitations. The feature and forward-label values agree with reconstruction from the supplied adjusted-price archive, and training-label maturity checks pass. That verification does not establish the historical point-in-time availability of vendor adjustment factors. The cohort also lacks a complete dated security master with delistings and symbol changes. The target is a clipped benchmark-relative log return, not a net portfolio payoff; trading costs, financing, capacity and compounded holdings are absent. Generalization to the full US equity universe and to executable trading decisions therefore remains untested.

Further confirmation would require a historically defined investable universe, a frozen analysis plan and new calendar information. The models and calibrated baseline would need to be compared on matched cross-sections, with sufficiently many market periods for dependence-aware inference. The same evaluation distinctions also arise wherever numerical forecasts are used to choose among simultaneous alternatives, but the size of the resulting ranking gap must be measured in each application.

11 Conclusion

An evaluation of return forecasting for equity selection must distinguish numerical forecast error from the ordering of securities available on the same date. In this fixed equity cohort, training-only calibration removes most of the apparent prediction-error advantage over raw historical Ret63. Residual-target ridge does not show a consistently supported rank improvement: its seven-, 21- and 63-session monthly block intervals all include zero. Repeated identity partitions and short calendar histories limit the precision of transfer claims.

The practical implication is an evaluation procedure: compare forecasts on a common target scale, retain directional rules as separate ranking baselines, examine within-date ranking directly and base uncertainty on the market calendar shared by repeated cross-sections. Its value here is to make an apparent forecasting gain and its limited decision relevance separately measurable.

A Proofs and Scope of the Mathematical Statements

A.1 Expected-reward identity sufficiency

Proof of Proposition 1. The candidatewise conditional expected reward of asset i given only (X_i, I_i) equals $\mathbb{E}[Y_i \mid X_i, I_i] = \mathbb{E}[Y_i \mid X_i]$ under the stated target law and its common state–outcome mapping. Thus within the restricted class of scores formed from each candidate's own information, replacing the identity-aware conditional-mean score by $g(X_i) = \mathbb{E}[Y_i \mid X_i]$ leaves all pairwise scores and their ordering unchanged, except for ties resolved independently of identity. This proof makes no claim about conditioning on the joint cross-section: in general $\mathbb{E}[Y_i \mid X_1, I_1, \ldots, X_n, I_n]$ need not equal $g(X_i)$. Nor does it assert that g can be estimated under arbitrary calendar shift or that means determine an objective involving higher moments or intertemporal holdings. □

A.2 Exact squared-error identity

Proof of Proposition 2. Set $a_t = \bar{Y}_t - \bar{s}_t$ and $e_{i,t} = R_{i,t} - Q_{i,t}$. By construction $\sum_i e_{i,t} = 0$. Expanding the square,

$$\frac{1}{n_t}\sum_i (Y_{i,t} - s_{i,t})^2 = \frac{1}{n_t}\sum_i (a_t + e_{i,t})^2 = a_t^2 + \frac{1}{n_t}\sum_i e_{i,t}^2,$$

because the cross term $2a_t n_t^{-1}\sum_i e_{i,t}$ vanishes. Multiplying by n_t/N and summing dates proves the pooled version. For date-averaged MSE one instead uses uniform date weights; these two summaries agree only with balanced cross-section sizes or special data configurations. □

A.3 A nondegenerate prediction–ranking counterexample

Proof of Theorem 1. Fix any cross-section size $n \geq 3$ and a date-common scalar m. Let $U_1, \ldots, U_n$ and $V_1, \ldots, V_n$ be independent standard normal random variables, with all $2n$ coordinates independent. For $\varepsilon > 0$ define outcomes $Y_i = m + \varepsilon U_i$ and predictors $S_i = m + \varepsilon V_i$. Every Y_i and S_i is continuous and predictions are not constant across candidates almost surely. Their expected per-candidate squared discrepancy is

$$\mathbb{E}[(Y_i - S_i)^2] = \varepsilon^2 \mathbb{E}[(U_i - V_i)^2] = 2\varepsilon^2,$$

which can be arbitrarily small. Because the two vectors of iid continuous coordinates are independent, their rank permutations are independent and uniform. Spearman's coefficient is their sample rank correlation and has expectation zero by permutation symmetry. Thus arbitrarily low expected squared loss can coexist with no expected rank information. Taking $\varepsilon = 0$ instead would produce ties and make Spearman undefined; the theorem uses $\varepsilon > 0$ throughout. □

Scope. The construction assumes the common component m is given to the predictor. This is an existence counterexample to a logical implication; it neither claims that real equity common returns are observable before they occur nor explains the dominant source of empirical squared error in Table 2. Its rank statement concerns an expectation over the constructed randomness, not zero Spearman in every realized sample.

A.4 Order invariance and universe changes

Proof of Proposition 3. Subtracting the same real number from both sides of a strict inequality preserves that inequality. The realized date mean $\bar{Y}_t$ is a common scalar. □

Proof of Lemma 1. Let $j \notin \mathcal{A}_t$ and suppose $s(X_{j,t}) > \max_{i \in \mathcal{A}_t} s(X_{i,t})$. The maximum after adjoining j is attained at j whereas the original maximum was attained in $\mathcal{A}_t$. Hence the maximizing action changes. □

A.5 What affine Ret63 calibration does to ranks

Suppose r is standardized in training as $z = (r - \mu_r)/\sigma_r$ and a ridge fit on z has intercept α and coefficient β. Its raw-unit form is $a + br$ with $b = \beta/\sigma_r$ and $a = \alpha - \beta\mu_r/\sigma_r$. For any non-tied pair $r_i \neq r_j$, the sign of $(a + br_i) - (a + br_j)$ is the sign of $b(r_i - r_j)$. If $b < 0$, the ranking is reversed. With finite ties the reversal is interpreted using consistent tie handling. The V2 code records the fitted raw-unit parameters for each partition; their signs are observations, not a rule selected on test outcomes.

B Supplementary Numerical Results

B.1 Full held-out model metrics

Table 8: Full $H = 7$ held-out summary from the frozen V2 verification output.

Model	MSE	MAE	Top-decile lift	Top-one regret
State ridge	0.004990	0.048595	−0.001180	0.168743
CS-residual ridge	0.005001	0.048617	−0.005485	0.161001
Ret63 calibrated	0.004958	0.048438	+0.000259	0.168380
Ret63 raw	0.044713	0.158117	+0.001390	0.158898

Table 9: Full $H = 21$ held-out summary from the frozen V2 verification output.

Model	MSE	MAE	Top-decile lift	Top-one regret
State ridge	0.010738	0.074811	−0.024084	0.274764
CS-residual ridge	0.010644	0.074280	−0.019733	0.252935
Ret63 calibrated	0.010628	0.074408	−0.006618	0.247482
Ret63 raw	0.050744	0.168057	−0.000996	0.238998

Table 10: Full $H = 63$ held-out summary from the frozen V2 verification output.

Model	MSE	MAE	Top-decile lift	Top-one regret
State ridge	0.029882	0.128523	−0.049458	0.422098
CS-residual ridge	0.029692	0.127762	−0.036118	0.405694
Ret63 calibrated	0.029388	0.127271	−0.010535	0.380978
Ret63 raw	0.073086	0.202453	−0.013068	0.397496

B.2 Additional inference details

Across the three horizons, the main rank comparison has 24 unique evaluation months and therefore nominal nonoverlapping-block counts eight, four and two at block lengths three, six and 12. The V2 output reports actual circular resampled-block counts 8, 4 and 2 per replicate. The use of circular blocks reduces endpoint artifacts but does not guarantee that 24 months support asymptotic resampling approximations. The code reports 5,000 bootstrap replicates and computes a mean-centered, two-sided exploratory p value with a $+1$ finite-replication correction. For block length 12, p values are intentionally left blank. The 21-session rank paired bootstrap at block lengths $3, 6, 12$ yields $p = 0.14577$, 0.18936, and no reported value; seven-session values are 0.41092, 0.30814, and blank; 63-session values are 0.93141, 0.89262, and blank. These provide no calendar-block basis for declaring positive rank improvement.

For the fixed-roster interaction, month-level seen/unseen differences are paired within dates and then near and future months are resampled separately. The near/future counts are 12 and 24 months. At block lengths three, six and 12, the near regime has four, two and one nominal nonoverlapping blocks, while the future regime has eight, four and two. Thus only the three-month results receive an exploratory two-sided p value. Even this shortest-block result depends on a short near period, the same single roster, and the use of an approximate percentile method. No familywise multiplicity correction is built into the inference output.

B.3 Archived horizon and universe checks

The original manuscript's five-split development and later 20-split results, its strict cutoffs and four frozen-universe subsets are historical outputs under the archival protocol. The V2 audit preserves rather than reruns these secondary analyses. The original universe table showed heterogeneity: the state-ridge Spearman values by frozen universe were about -0.01170, -0.01928, $+0.01699$, and $+0.06320$, while CS-residual ridge values were -0.02646, -0.02493, $+0.01388$, and $+0.05817$. These descriptive differences corroborate sensitivity to the defined candidate population, not independence of universe composition. The publication-grade interpretation of such subsets remains restricted by their historical selection and price-data conventions.

B.4 Historical evidence and V2 evidence are not exchangeable

The archival 20-identity-split mean Spearman difference is $+0.003656$, with an interval computed at the level used in the original study. V2 estimates $+0.040003$ at 21 sessions under a calendar-frozen 2021–22 evaluation and computes a three-month interval spanning zero. Because the windows and eligibility structures differ, these are distinct contrasts. An analysis using only one of the intervals while treating both point estimates as identical would misstate the evidence. The historical estimates are retained as archival results. The V2 verification CSVs document a separate calculation, not new independent market replications.

C Reproducibility and Data Provenance

C.1 Numerical analysis and reporting materials

The numerical analysis code consumes archived asset–month panels, fits the specified models and exports the evaluation statistics. Frozen CSV outputs for $H \in \{7, 21, 63\}$ accompany the reporting materials, along with figure-generation scripts and the manuscript source. Regenerating

figures from the reported CSVs reproduces the presentation of the results; repeating the model estimation also requires the separate numerical-analysis code and underlying market inputs.

The main configuration fixes seed 20261020, ridge penalty 5, 20 identity partitions, a 20% test-identity fraction, a training cutoff of 2020-12-31, and a 2021-01-04 to 2022-12-30 evaluation window. Calendar-block analyses use $B = 5{,}000$ replicates and block lengths of 3, 6 and 12 months. The time–identity diagnostic fits using data through 2019-12-31 and compares a 12-month near period with a 24-month future period. This analysis was constructed after the preceding identity-holdout results were examined.

C.2 Frozen verification records

Each of the three horizons has the following exported summaries: `heldout_summary.csv`, `dependence_robust_intervals.csv`, `factorial_summary.csv`, `factorial_interaction_spearman.csv`, `ret63_calibration_by_split.csv`, and `ret63_calibration_factorial.csv`. These outputs provide the numerical basis for the reported tables and plots. They are generated under the model and cohort restrictions in Section 6.

C.3 Point-in-time limits

Public vendor prices may be revised retrospectively, including their adjustment for corporate actions. Matching historical features and labels to one archived price version verifies computation within that version; it does not establish when the vendor supplied each adjusted value. The analysis also lacks a complete dated universe with all delisted constituents and identity changes. The fixed-cohort results should be interpreted under those data limitations.

Data and Code Availability

The numerical analysis uses archived equity panels and the procedures described in Section 6 and Appendix C. The public code and results repository is `Prediction-Ranking-Transfer-Audit`. It provides the evaluation implementation, configurations, automated tests, and selected frozen numerical output files for all three horizons. The accompanying LaTeX source package includes the manuscript, figure-generation scripts, and verification tables. Complete re-estimation of market results requires compatible underlying price data obtained separately; third-party historical market data are not redistributed in the repository because their reuse may be subject to vendor terms.

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Disclosure of interest

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References

Arnott, Robert D., Campbell R. Harvey, and Harry Markowitz (2019). "A Backtesting Protocol in the Era of Machine Learning". In: *The Journal of Financial Data Science* 1.1, pp. 64–74. DOI: 10.3905/jfds.2019.1.064.

Bailey, David H., Jonathan M. Borwein, Marcos López de Prado, and Qiji Jim Zhu (2014). "Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance". In: *Notices of the American Mathematical Society* 61.5, pp. 458–471. DOI: 10.1090/noti1105.

DeMiguel, Victor, Lorenzo Garlappi, and Raman Uppal (2009). "Optimal Versus Naive Diversification: How Inefficient is the 1/N Portfolio Strategy?" In: *The Review of Financial Studies* 22.5, pp. 1915–1953. DOI: 10.1093/rfs/hhm075.

Elmachtoub, Adam N. and Paul Grigas (2022). "Smart "Predict, then Optimize"". In: *Management Science* 68.1, pp. 9–26. DOI: 10.1287/mnsc.2020.3922.

Gu, Shihao, Bryan Kelly, and Dacheng Xiu (2020). "Empirical Asset Pricing via Machine Learning". In: *The Review of Financial Studies* 33.5, pp. 2223–2273. DOI: 10.1093/rfs/hhaa009.

Harvey, Campbell R., Yan Liu, and Heqing Zhu (2016). "... and the Cross-Section of Expected Returns". In: *The Review of Financial Studies* 29.1, pp. 5–68. DOI: 10.1093/rfs/hhv059.

Künsch, Hans R. (1989). "The Jackknife and the Bootstrap for General Stationary Observations". In: *The Annals of Statistics* 17.3, pp. 1217–1241. DOI: 10.1214/aos/1176347265.

Poh, Daniel, Bryan Lim, Stefan Zohren, and Stephen Roberts (2021). "Building Cross-Sectional Systematic Strategies by Learning to Rank". In: *The Journal of Financial Data Science* 3.2, pp. 70–86. DOI: 10.3905/jfds.2021.1.060.

Politis, Dimitris N. and Joseph P. Romano (1994). "The Stationary Bootstrap". In: *Journal of the American Statistical Association* 89.428, pp. 1303–1313. DOI: 10.1080/01621459.1994.10476870.

White, Halbert (2000). "A Reality Check for Data Snooping". In: *Econometrica* 68.5, pp. 1097–1126. DOI: 10.1111/1468-0262.00152.